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ORIGINAL ARTICLE

Systolic Blood Pressure Targets in Patients With Hypertension With and Without Cardiovascular Disease: A Pooled Analysis of 4 Randomized Clinical Trials

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BACKGROUND: Inconsistent selection of reference groups may contribute to inconclusive blood pressure control targets for patients with hypertension. This study aims to identify optimal systolic blood pressure (SBP) targets for individuals with and without cardiovascular disease (CVD) by employing a novel metric combining blood pressure magnitude and duration.

METHODS: We conducted a secondary analysis of individual-level data from 4 clinical trials (ACCORD, ALLHAT, SHEP, and SPRINT). Participants aged >80 years, with prevalent heart failure, or with missing follow-up SBP measurements or CVD history were excluded. The proportion of time-in-target range (TTR) was estimated using linear interpolation to quantify longitudinal SBP changes across predefined SBP ranges. Associations between TTR and major adverse cardiovascular events were assessed using stratified Cox proportional hazards models, analyzed separately by CVD history, and stratified by study trials, with hazard ratios (HRs) estimated per 10% increase in TTR. Lower TTR within the same target served as the implicit reference.

RESULTS: Among 59 727 participants included in the analysis, the mean age was 65.8 (SD, 6.9) years, 52.3% were men, and 24 524 (41.1%) had a history of CVD. Higher TTR within SBP ranges of 110 to 140 mm Hg was consistently associated with a lower risk of major adverse cardiovascular events in participants with and without CVD. For SBP of 110 to 120 mm Hg, HRs were 0.89 (95% CI, 0.86–0.92) in participants without CVD and 0.93 (95% CI, 0.90–0.95) among those with CVD; corresponding HRs were 0.86 (95% CI, 0.83–0.88) and 0.90 (95% CI, 0.88–0.92) for 120 to 130 mm Hg, and 0.94 (95% CI, 0.92–0.96) and 0.93 (95% CI, 0.91–0.95) for 130 to 140 mm Hg. SBP ≥150 mm Hg was associated with increased major adverse cardiovascular event risk in both groups, whereas SBP <110 mm Hg was associated with reduced risk only in participants without CVD (HR, 0.91 [95% CI, 0.86–0.97]) and not among those with CVD (HR, 1.02 [95% CI, 0.98–1.06]).

CONCLUSIONS: Restricted cubic spline analyses demonstrated a J-shaped association between achieved SBP and major adverse cardiovascular event irrespective of CVD history. Maintaining SBP within 110 to 140 mm Hg was similarly associated with lower cardiovascular risk in patients with and without CVD. Risk appeared lowest when SBP was between 120 and 130 mm Hg, while an SBP target <110 mm Hg may offer additional benefit among patients without CVD.

Key Words: blood pressure ■ cardiovascular diseases ■ heart disease risk factors ■ hypertension ■ risk factors

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WHAT IS KNOWN

- Achieved systolic blood pressure (SBP) level is strongly associated with cardiovascular risk, and time-in-target range reflects both blood pressure magnitude and duration.
- Whether optimal SBP targets differ between individuals with and without cardiovascular disease remains uncertain.

WHAT THE STUDY ADDS

- Similar optimal SBP for both patients with hypertension with and without cardiovascular disease when maintaining SBP within the range of 110 to 140 and 120 to 130 mm Hg target appeared most favorable.
- Benefit may only be offered in patients without cardiovascular disease when SBP control targets <110 mm Hg.

Nonstandard Abbreviations and Acronyms

BP	blood pressure
CVD	cardiovascular disease
HDL	high-density lipoprotein
HR	hazard ratio
MACE	major adverse cardiovascular event
MI	myocardial infarction
RCS	restricted cubic spline
SBP	systolic blood pressure
TTR	time-in-target range

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Current hypertension guidelines provide differentiated blood pressure (BP) targets for patients with and without cardiovascular disease (CVD).[1–4] Key areas of controversy include whether patients with and without CVD should share the same systolic BP (SBP) goal, and whether further reduction below 120 mm Hg confers additional benefit. The 2024 European Society of Cardiology[1] recommended 120 to 129 mm Hg for both patients with and without known CVD. The 2024 European Society of Hypertension guidelines also recommended a target SBP <130 mm Hg in most patients, but mentioned benefit of BP lowering below 120 mm Hg is unclear.[5] However, the 2021 World Health Organization guideline recommends SBP <140 mm Hg for those without comorbidities and <130 mm Hg for patients with CVD or high cardiovascular risk.[4] Although these recommendations have evolved with accumulating evidence, the optimal SBP target for patients with or without CVD and the lower target threshold in patients with hypertension remains inconclusive.

Beyond the randomized clinical trials focused on BP targets,[6,7] several studies have endeavored to explore the SBP target in individuals with and without established CVD.[8–10] However, these studies predominantly relied on single-point measurements or mean achieved SBP,[9,10] which may not adequately capture the dynamic and fluctuating nature of BP control over time.[11] In addition, these studies used reference groups with established clinical benefits, such as 120 to 129 or 120 to 140 mm Hg, which may introduce inconsistencies and subjectivity in the selection of the comparator.[9,10] As a result, they may underestimate the CVD risk associated with BP lowering and obscure potential harms from overly aggressive targets.[9,10]

To address this limitation, our previous research,[12] along with numerous studies, has proposed the use of the time-in-target range (TTR) metric,[13–16] which quantifies the proportion of time a patient's BP remains within a predefined target range. This approach integrates the magnitude and duration of BP control into a single comprehensive measure, offering a more nuanced assessment of treatment efficacy. Building on this, the present study investigates the relationship between TTR and major adverse cardiovascular event (MACE) across comprehensive BP-lowering targets in patients with and without CVD, aiming to determine whether the optimal SBP control range differs by CVD history.

METHODS

Data Sources and Study Population

All data and materials have been made publicly available at the National Heart, Lung, and Blood Institute Biologic Specimen and Data Repository Information Coordinating Center and can be accessed at http://biolincc.nhlbi.nih.gov/home/.

This study was a secondary analysis using participant-level data from 4 major clinical trials: ACCORD (n=10 251), ALLHAT (n=42 418), SHEP (n=4 736), and SPRINT (n=9 361). The design and primary findings of these trials have been published.[6,17–21] A detailed description of the study design and population is provided in the Supplemental Methods.

Data were harmonized and pooled to facilitate comparability and consistency. For our analysis, we excluded participants aged >80 years or with a history of heart failure, as these populations have distinct BP management strategies, higher baseline cardiovascular risk, and different responses to antihypertensive therapy, which could confound the study outcomes.[22,23] Additional exclusion criteria included missing data or the occurrence of the primary event before the first follow-up. Further details regarding the exclusion criteria are presented in Figure 1. All studies received ethical approval from their participating institutions. CVD history was defined according to the original criteria of each trial. SPRINT included both clinical and subclinical CVD, while ACCORD used its predefined CVD history. In ALLHAT and SHEP, CVD history was defined as prior myocardial infarction (MI) or stroke, history of coronary angioplasty or other revascularization, and ECG changes. Detailed definitions are provided in the Supplemental Methods.

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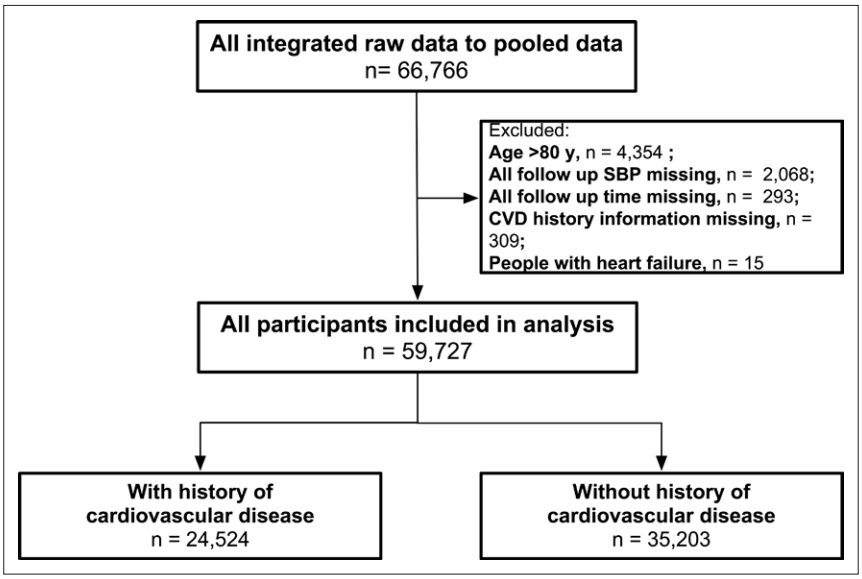

Figure 1. Research flowchart and the exclusion criteria.
In total, 66 766 individual-level patient records were integrated into pooled data. After applying exclusion criteria, a total of 7039 participants were excluded: 4354 were aged >80 years, 15 had a history of heart failure, 2068 had missing follow-up systolic blood pressure (SBP), 293 lacked follow-up data, and 309 had missing information on cardiovascular disease (CVD) history. The final analytic sample included 59 727 participants, who were further categorized into 2 groups: those with a history of CVD (n=24 524) and those without (n=35 203).

This study used fully deidentified secondary data and was granted a waiver of ethical approval by the Ethical Review Board of the First Affiliated Hospital of Xi'an Jiaotong University (MC-KYLLSL-2023-005).

BP Measurements

BP measurements in this study were conducted following standardized protocols across the 4 clinical trials, using both automated and manual sphygmomanometers. In SPRINT[24] and ACCORD,[17] BP was measured using automated Omron devices, with multiple readings averaged for analysis. The ALLHAT[25] trial utilized mercury sphygmomanometers, recording the average of 2 seated measurements taken 30 seconds apart. In SHEP,[26] BP was measured by trained technicians using random-zero sphygmomanometers, with SBP calculated as the average of 2 readings. Detailed measurement procedures for each trial are provided in the Supplemental Methods.

Study Outcomes

In the present study, the primary outcome was the incidence of the first MACE, defined as the occurrence of at least one of the following: MI, stroke, or CVD mortality. Our secondary outcomes included MI, stroke, and CVD mortality. Outcomes were defined and ascertained according to the criteria established in each parent study.[17,27–29]

TTR Calculation

TTR was defined as the percentage of time that a participant's SBP remained within a specified target range over a certain period (Figure S1).[30] SBP TTR was estimated using linear interpolation, which assumes a linear trend between 2 consecutive SBP measurements.[31] We initially applied broader categories based on clinically relevant thresholds commonly referenced in hypertension guidelines with 110, 120, 130, and 140 mm Hg.[1–5,32] Then, we used narrow SBP intervals in 5-mm Hg increments from 90 to 160 mm Hg, including boundary categories for SBP <90 and ≥160 mm Hg. This dual-categorization approach enabled a comparative analysis between fine-grained SBP control and guideline-aligned target ranges.

Statistical Analysis

Baseline characteristics of the study population were presented as mean±SD for continuous variables and counts with percentages for categorical variables. Differences between groups were assessed using the χ^2 test for categorical variables and the Wilcoxon rank-sum test for continuous variables.

To assess potential nonlinear relationships between average visit SBP and the incidence rate of MACE, restricted cubic spline (RCS) analysis within Poisson regression models was used. The average visit SBP was defined as the mean of all SBP measurements recorded during follow-up visits. RCS models were adjusted for key cardiovascular risk factors, including age, gender, race, body mass index (BMI), baseline SBP, smoking status, diabetes status, and study cohort. The number and placement of knots (5%, 35%, 65%, and 95%) were determined based on the percentile distribution of the data and guided by previous research.[33] For potential nonlinearity in the association between TTR and MACE, we applied RCS functions within Poisson regression models for each SBP target. Because event numbers were limited in certain strata, spline models were fitted using 3 knots, and all models were adjusted for key cardiovascular risk factors as RCS models of average visit SBP.

Associations between SBP TTR and MACE were evaluated separately in participants with and without CVD using a stratified Cox proportional hazards model. TTR was calculated independently within each SBP target, ranging from 0% to 100%. The hazard ratio (HR) per 10% TTR reflects the risk change associated with spending 10% more time within the same SBP interval. The model was adjusted for age, gender, race, BMI, baseline SBP, smoking status, diabetes status, and stratified by study cohort. To assess whether any single trial exerted undue influence on the overall findings, we performed a leave-one-out sensitivity analysis by sequentially excluding each trial from the analysis. Furthermore, the associations between SBP TTR and each component of MACE (MI, stroke, and CVD mortality) were also analyzed. To ensure model validity, we tested the proportional hazards assumption using Schoenfeld residuals. We further adjusted for diastolic BP, total and HDL (high-density lipoprotein)-cholesterol, and estimated glomerular filtration rate

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to assess potential residual confounding. To assess potential competing-risk bias, we fitted Fine-Gray subdistribution hazard models treating non-CVD mortality as a competing event. To further evaluate potential effect modification by CVD history, a pooled Cox model including a TTR×CVD interaction term was also fitted as a sensitivity analysis. HRs and 95% CIs were estimated for all models. All statistical analyses were conducted using R software (version 4.4.2; R Foundation for Statistical Computing).

RESULTS

We integrated 66 766 individual-level patient records from 4 clinical trials into a pooled data set: ACCORD (n=10 251), ALLHAT (n=42 418), SHEP (n=4736), and SPRINT (n=9361). After applying exclusion criteria, a total of 7039 participants were excluded: 4354 were aged >80 years, 15 had a history of heart failure, 2068 had missing follow-up SBP, 293 lacked follow-up data, and 309 had missing information on CVD history. The final analytic sample included 59 727 participants, who were further categorized into 2 groups: those with a history of CVD (n=24 524) and those without (n=35 203). Further details on participant selection are presented in Figure 1. The mean baseline age of included patients was 65.82±6.87 years, with 52.32% being male and 62.02% identifying as white. Participants with a history of CVD were more likely to experience outcomes compared with those without CVD (13.26% versus 7.47%). The study involved an average of 13.77±7.10× of SBP measurements per individual during the study period (Table; Table S1).

In the RCS analysis, both groups showed a similar modest increase pattern at higher average visit SBP in MACE incidence, but diverged slightly at lower, indicating a J-shaped curve regardless of CVD history (Figure 2A). Among participants without CVD history, MACE incidence increased modestly as SBP rose. For those with CVD, the lowest incidence occurred within 120 to 130 mm Hg. Below this range, the MACE incidence rate increased as average SBP decreased, while, above this range, the rate also rose with increasing average SBP (Figure 2A). RCS analyses also revealed clear nonlinear associations between TTR and MACE across all SBP targets (Figure S2).

This trend was also similarly evident in the stratified Cox proportional hazards models analyzing the association between TTR and MACE across various SBP target ranges in patients with and without CVD. Within the 110- to 120-mm Hg range, the HRs for MACE were 0.89 (95% CI, 0.86–0.92) in patients without CVD and 0.93 (95% CI, 0.90–0.95) in those with CVD. For 120 to 130 mm Hg, the corresponding HRs were 0.86 (95% CI, 0.83–0.88) and 0.90 (95% CI, 0.88–0.92); for 130 to 140 mm Hg, 0.94 (95% CI, 0.92–0.96) and 0.93 (95% CI, 0.91–0.95); and for 140 to 150 mm Hg, 1.06 (95% CI, 1.03–1.08) and 1.06 (95% CI, 1.04–1.08), respectively. When SBP targets were ≥150 mm Hg, higher TTR values were associated with increased MACE risk in both groups. The HR for patients without CVD was 1.13

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Table. Characteristics of the Participants at Baseline

Characteristic	Overall	Without CVD	With CVD	*P* value
No. of participants	59 727	35 203 (58.94%)	24 524 (41.06%)	<0.001
Age, y; mean (SD)	65.82 (6.87)	65.33 (6.76)	66.53 (6.96)	<0.001
Gender, n (%)				<0.001
Male	31 285 (52.32%)	17 011 (48.30%)	14 274 (58.23%)	
Female	28 442 (47.68%)	18 192 (51.70%)	410 250 (41.77%)	
Race, n (%)				<0.001
White	37 030 (62.02%)	20 992 (60.00%)	16 038 (65.00%)	
Other races	22 697 (37.98%)	14 211 (40.00%)	8486 (35.00%)	
Baseline blood pressure, mm Hg; mean (SD)				
Systolic	145.02 (17.37)	145.62 (17.78)	144.16 (16.72)	<0.001
Diastolic	81.48 (11.01)	81.39 (10.96)	81.62 (11.07)	<0.001
Baseline BMI, kg/m^2; mean (SD)	30.17 (6.09)	30.45 (6.35)	29.78 (5.68)	<0.001
Number of SBP measurements per participant, mean (SD)*	13.77 (7.10)	14.64 (7.36)	12.53 (6.53)	<0.001
Research, n (%)				<0.001
ACCORD	10 066 (16.85%)	6546 (18.60%)	3520 (14.35%)	
ALLHAT	37 636 (63.03%)	18 387 (52.23%)	19 249 (78.50%)	
SHEP	4001 (6.70%)	3749 (10.65%)	252 (1.03%)	
SPRINT	8070 (13.51%)	6538 (18.57%)	1532 (6.25%)	

BMI indicates body mass index; CVD, cardiovascular disease; and SBP, systolic blood pressure.
*SBP measurement times per individual=total SBP measurement times/number of participants.

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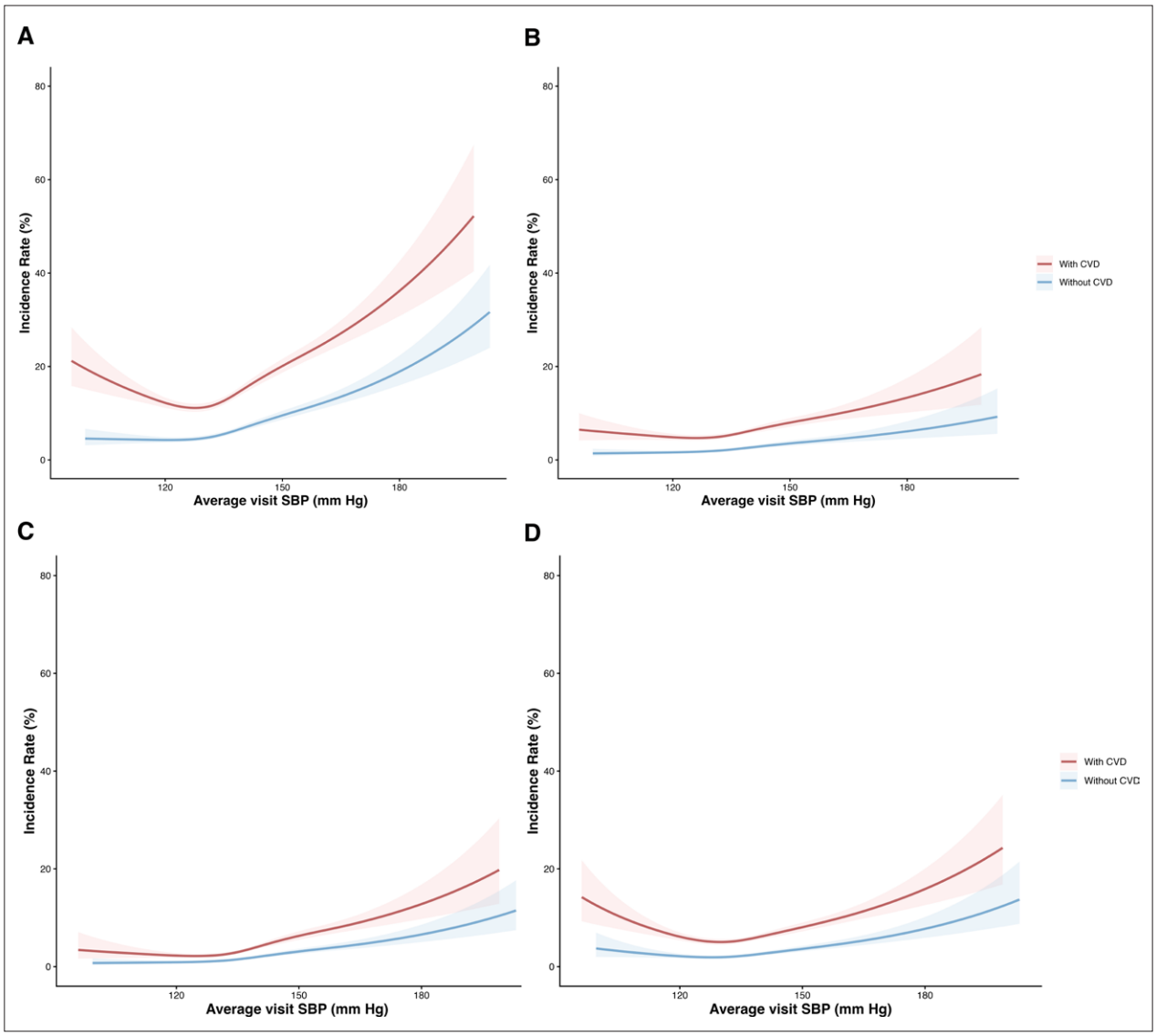

Figure 2. Association between average visit systolic blood pressure (SBP) and incidence rate.
Restricted cubic spline analysis shows the relationship between average visit SBP and incidence rate of major adverse cardiovascular event (MACE; **A**), myocardial infarction (MI; **B**), stroke (**C**), and cardiovascular disease (CVD) mortality (**D**) in patients with hypertension with and without CVD history.

(95% CI, 1.11–1.15), and for those with CVD, the HR was 1.11 (95% CI, 1.09–1.13; Table S2). These trends were consistently observed across leave-one-out sensitivity analysis (Figures S3).

A similar pattern was observed when SBP targets were analyzed in finer 5-mm Hg increments across the 90- to 160-mm Hg range (Figure S4; Table S3). A similar protective association within the 110- to 140-mm Hg range was observed in both participants with and without CVD. For SBP levels above 140 mm Hg, both groups of people showed a consistent pattern of increased risk. For SBP levels below 110 mm Hg, a divergence in trends emerged, with differing associations observed between the CVD and non-CVD subgroups.

Similar patterns were observed when MI and CVD mortality were used as the end points in both RCS and stratified Cox proportional hazards models (Figures 2B, 2D, 3B, and 3D). Within the SBP target range of 110 to 140 mm Hg, higher TTR values were consistently associated with a lower risk of MI, stroke, and CVD mortality across both groups, supporting the protective effect of this target range. At SBP targets <110 mm Hg, higher TTR was associated with reduced MI risk in patients without CVD (HR, 0.84 [95% CI, 0.76–0.92]), but the association was weaker and nonsignificant in those with CVD (HR, 1.01 [95% CI, 0.96–1.07]; Figure 3B; Table S2). A similar trend was observed for CVD mortality (with CVD: HR, 1.09 [95% CI, 1.04–1.15]; without CVD: HR, 1.04 [95% CI, 0.95–1.14]; Figure 3D; Table S2). In contrast, stroke risk decreased with higher TTR in both groups across all SBP targets <140 mm Hg (Figures 2C and 3C; Table S2).

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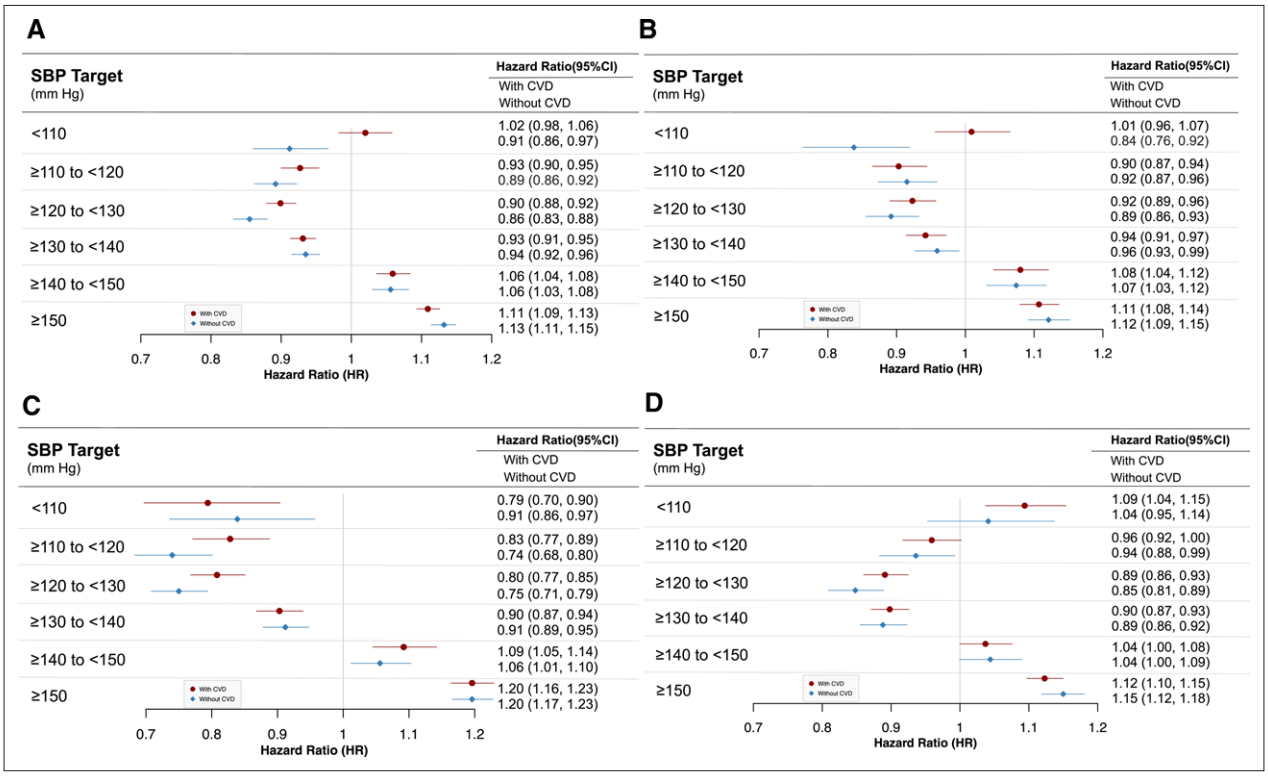

Figure 3. Association between time-in-target range (TTR) and cardiovascular outcomes.
Stratified Cox proportional hazards models illustrate the association between TTR of systolic blood pressure (SBP) and risk of major adverse cardiovascular event (MACE; **A**), myocardial infarction (MI; **B**), stroke (**C**), and cardiovascular disease (CVD) mortality (**D**) in patients with hypertension with and without a history of CVD. TTR was calculated per the SBP target range. Hazard ratios (HRs) for participants with and without a history of CVD were obtained from separate stratified Cox models in each SBP target and represent the change in risk per 10% increase in TTR. For **A** through **D**, the numbers of participants and events are given as follows (with CVD vs without CVD): MACE, 3251/24 524 (13.26%) vs 2630/35 203 (7.47%); MI, 1342/24 524 (5.47%) vs 1053/35 203 (2.99%); stroke, 853/24 524 (3.48%) vs 873/35 203 (2.48%); and cardiovascular mortality, 1438/24 524 (5.86%) vs 920/35 203 (2.61%).

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Sensitivity analyses adjusting for additional cardiovascular risk factors, including diastolic BP, total and HDL-cholesterol, and estimated glomerular filtration rate, yielded results that were highly consistent with the primary models. HRs for all SBP targets remained directionally unchanged and of similar magnitude (Table S4; Figure S5). Competing-risk analyses using Fine-Gray models yielded effect estimates similar to the primary Cox models, indicating that non-CVD mortality did not materially influence the observed associations (Table S5). Formal interaction testing supported that the relationship between TTR and MACE differed between those with and without CVD for most SBP ranges (Table S6), with significant TTR and CVD interaction terms for all targets except 130 to 140 mm Hg.

DISCUSSION

This study conducted an integrated analysis using data from 4 large-scale clinical trials by utilizing the TTR method to identify the optimal BP control targets in patients with hypertension with or without CVD. The findings suggest that maintaining SBP within 110 to 140 mm Hg was similarly associated with reduced MACE risk in patients with CVD or without CVD. Risk appeared lowest when SBP was between 120 and 130 mm Hg, while additional benefit at SBP <110 mm Hg may be observed among patients without CVD.

Optimal SBP targets remain a central challenge in hypertension management. Current guidelines provide differentiated recommendations based on patient characteristics such as age and comorbid conditions. For the general hypertensive population, some recent guidelines converge on a target SBP of <130 mm Hg though recommendations regarding the lower limit remain variable. The 2017 American Heart Association BP guideline suggests that SBP <130 mm Hg may be reasonable,[2] while the 2020 ISH guideline recommends lowering SBP to <130 mm Hg (but >120 mm Hg) for individuals aged <65 years and <140 mm Hg for those aged ≥65 years.[32] The 2021 World Health Organization guideline advises an SBP target of <140 mm Hg.[4] The 2024 European Society of Hypertension guidelines propose a target SBP <130 mm Hg in most patients up to 79 years old, and the mentioned benefit-to-risk ratio of SBP <120 mm Hg is unclear,[5] whereas the 2024 European Society of Cardiology guidelines recommend a more specific target of 120 to 129 mm Hg for most patients.[1]

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For patients with established CVD, most guidelines favor more conservative SBP targets. The 2017 American Heart Association, 2020 ISH, and 2021 World Health Organization guidelines all endorse <130 mm Hg.[2,4,32] The 2024 European Society of Cardiology suggests 120 to 129 mm Hg as the target range and highlights 120 mm Hg as a potential optimal level if well tolerated.[1] Despite these recommendations, the optimal SBP target, including whether a lower boundary should be specified, remains uncertain. This lack of consensus partly reflects the limited evidence on whether SBP targets should vary based on CVD history.

The variation in guideline-recommended SBP targets largely stems from the heterogeneity of evidence across randomized controlled trials.[6,7,27,34–36] Clinical trials have varied in design, and some trials assessed the effect of treatment versus no treatment, while others examined different treatment intensities or targeted SBP levels. These variations, along with reliance on average or single-timepoint BP measures, limit direct comparisons and may partly explain the inconsistent conclusions regarding optimal BP levels. For patients with CVD, post hoc analyses by Böhm et al[9] and Vidal-Petiot et al[10] found that SBP <120 mm Hg was linked to increased risks of adverse cardiovascular outcomes in patients with prior CVD. However, these studies primarily used mean SBP values and predefined reference groups (eg, 120–129 or 120–140 mm Hg), which may potentially underestimate cardiovascular risk in the comparison groups and overlook the harms associated with excessive BP lowering. In our analysis, TTR addresses these limitations by providing a more dynamic assessment of BP control over time and has been shown to predict cardiovascular outcomes independently of mean achieved SBP.[30] TTR captures both the level of SBP and the duration for which it is maintained, providing a continuous exposure-response interpretation. Thus, a per 10% TTR estimate represents the risk gradient associated with more sustained BP control within a specific target range, offering complementary insight beyond traditional BP metrics. We found that maintaining SBP within the range of 110 to 140 mm Hg was similarly associated with reduced MACE risk in both patients with and without a history of CVD. Among these, the 120- to 130-mm Hg interval may be associated with the most favorable outcome. Notably, when SBP <110 mm Hg, a protective effect was evident only among patients without CVD. These associations remained consistent across multiple sensitivity analyses, reinforcing the robustness of our results. These findings suggest that the optimal SBP target range is largely shared among patients with or without CVD. Individual risk profiles may warrant caution when considering more intensive BP lowering.

To better understand the consistency of associations across outcomes, we separately examined the individual components of MACE, including MI, stroke, and CVD mortality. Both MI and CVD mortality exhibited trends similar to the overall MACE outcome. However, stroke risk declined progressively with higher TTR values <140 mm Hg, regardless of CVD history, highlighting a distinct pattern compared with other MACE components. This finding aligns with previous studies, such as the ACCOMPLISH trial[37] and the VALUE trial,[38] which reported a reduced stroke incidence when SBP was lowered to a relatively lower target. The divergent trend may be attributed to the unique pathophysiology of stroke. Hemorrhagic stroke, in particular, is highly sensitive to BP fluctuations. Chronic hypertension weakens cerebral small vessels, increasing the risk of rupture and hemorrhage,[39] whereas lower SBP reduces vascular stress and, thereby, decreases hemorrhagic risk. Furthermore, in ischemic stroke, aggressive SBP control may help prevent artery-to-artery embolism and reduce the incidence of lacunar infarcts caused by small-vessel disease. In contrast, MI and CVD mortality are more closely linked to coronary artery perfusion, which may explain their differential response to intensive BP lowering.

TTR also has potential clinical utility as an intuitive metric of long-term BP control quality. TTR does not require continuous or daily BP measurements; rather, it benefits from having multiple readings over time to capture the longitudinal pattern of BP control. Existing TTR studies, including those based on SPRINT[30] and ACCORD,[40] have relied on routine clinic BP collected every 1 to 3 months, demonstrating that intermittent measurements are sufficient for estimating TTR. Nevertheless, higher-frequency monitoring may further improve its precision. Ambulatory BP monitoring provides a complete 24-hour profile and may better capture nocturnal BP and morning surges, whereas home BP monitoring represents a practical and scalable approach that can be readily integrated into digital hypertension management platforms. As home BP monitoring continues to expand in clinical practice, TTR may evolve into a clinically useful and patient-friendly metric for tracking BP control and guiding treatment adjustments in both real-world and remote-monitoring settings.

This study has several limitations. First, as a post hoc analysis, it is inherently subject to potential reverse causality, which cannot be entirely ruled out. While we examined the relationship between TTR metrics and cardiovascular outcomes, it remains possible that preexisting health conditions or underlying disease progression influenced both BP regulation and clinical outcomes, rather than TTR itself being the primary determinant. Future prospective studies are needed to establish a clearer causal relationship. Second, the data utilized in this study were derived from 4 clinical trials, with rigorous BP management protocols. These controlled conditions may have reduced the variability between TTR and average BP, potentially influencing the observed associations. Consequently, our findings may not fully reflect

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real-world BP management, where patient adherence and treatment intensity vary. Third, this study did not impose restrictions on the duration of BP control, as our primary objective was to explore broad trends rather than assess the effects of specific control timeframes on outcomes. Future research should investigate how different durations of BP control impact clinical outcomes, which may provide further insights into optimizing long-term BP management strategies. In addition, potential adverse effects commonly associated with low BP levels (eg, symptomatic hypotension, electrolyte imbalance, and frailty) could not be evaluated, as these outcomes were not consistently collected or defined across the included trials.

Cardiovascular benefits were consistently associated with maintaining SBP within the range of 110 to 140 mm Hg, in both patients with hypertension with and without CVD, suggesting a similar optimal SBP target for both populations. Among these, the 120- to 130-mm Hg target appeared most favorable. While when SBP targets <110 mm Hg, additional benefit may only be offered in patients without CVD.

ARTICLE INFORMATION

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Disclosures

None.

Supplemental Material

Supplemental Methods
Tables S1–S6
Figures S1–S5
References 6, 17, 19, 24, 25, 26, 27, 28, 29

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